

Alessio Amaolo

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EDUCATION

- Princeton University** – Ph.D. Chemistry and Materials. Advised in Electrical and Computer Engineering. Expected 2027
- Princeton University (3.9/4.0 GPA)** – M.A. Chemistry (Pass with distinction) May 2023
- Selected coursework: Mathematical Methods of Engineering, Numerical Algorithms for Scientific Computing (Python, C++), Nanophotonics (MPB/Scheme), Quantum Optics, Condensed Matter Physics, Quantum Field Theory
- California Institute of Technology (3.9/4.0 GPA)** BSc Chemistry June 2021
- Selected graduate coursework: Computational Physics (Python), Quantum Computation, Advanced Condensed Matter Physics, Advanced Inorganic Chemistry
 - Selected undergraduate coursework: Computing Systems (C, Assembly), Programming Methods (C++), Bayesian Statistics (R)

PUBLICATIONS – [Google Scholar](#) (* indicates equal contribution)

13. Meiburg, A., *et al.* (including **Amaolo, A.**). The Bound-Beating Benchmark: A continuous, lean-verified benchmark of open quantitative problems. In preparation.
12. **Amaolo, A.**, Molesky, S., Rodriguez, A.W., Johnson, S.G., Chao, P. The photonic gearbox: ultra-high Purcell enhancement via symmetry breaking. In preparation.
11. Chen, F.J., **Amaolo, A.**, Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. [Optically Incoherent Photonic Mutual Information](#). *arXiv:2607.13153*. 2026.
10. **Amaolo, A.**, Chen, F., Chao, P., Maldonado, T.J., Strekha, B., Molesky, S., Rodriguez, A.W. Optimal inverse design in nanophotonics: a tutorial. *Advances in Optics and Photonics (Invited review paper)*. In preparation.
9. Virally, P., Chao, P., **Amaolo, A.**, Rodriguez, A.W., Molesky, S. [How many channels can a photonic system support?](#) *arXiv:2510.01128*. 2025.
8. Molesky, S., Chao, P., **Amaolo, A.**, Rodriguez, A.W. [Inferring structure via duality for photonic inverse design](#). *arXiv:2504.14083*. 2025.
7. Chao, P.*, **Amaolo, A.***, Molesky, S., Rodriguez, A.W. [Bounds as blueprints: towards optimal and accelerated photonic inverse design](#). *Optics Express*. 2026.
6. **Amaolo, A.***, Chao, P.*, Strekha, B., Clarke, S., Mohajan, J., Molesky, S., Rodriguez, A.W. [Maximum Shannon capacity of photonic structures](#). *npj Nanophotonics*. 2026.
5. Strekha, B., **Amaolo, A.**, Mohajan, J., Chao, P., Molesky, S., Rodriguez, A.W. [Limitations on bandwidth-integrated passive cloaking](#). *Physical Review A*. 2024.
4. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. [Physical limits on Raman scattering: the critical role of pump and signal co-design](#). *Physical Review A Letters*. 2024.
3. Maldonado, T.J., Pham, D.N., **Amaolo, A.**, Rodriguez, A.W., Türeci, H.E. [Negative electrohydrostatic pressure between superconducting bodies](#). *Physical Review B*. 2024.
2. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. [Can photonic heterostructures provably outperform single-material geometries?](#) *Nanophotonics*. 2024.
1. McNicholas, B.J., Nie, C., Jose, A., Oyala, P.H., Takase, M.K., Henling, L.M., Barth, A.T., **Amaolo, A.**, Hadt, R.G., Solomon, E.I., Winkler, J.R., Gray, H.B., Despagnet-Ayoub, E. [Boronated Cyanometallates](#). *Inorg. Chem.* 2022.

RESEARCH EXPERIENCE

Student Researcher, Google DeepMind

Research on computational lithography and mask process correction for chip fabrication optimization.

Summer
2025

Graduate Researcher in Rodriguez Group, Princeton

Performing research at the intersection of convex optimization and photonics focusing on calculating fundamental limits on photonic device performance.

2021-
Present

- Developed novel algorithms for photonic inverse design utilizing convex relaxations, beating state-of-the-art algorithms 10x in performance in 10x fewer iterations.
- Formulated first information-theoretic treatment of photonic devices to calculate optimal electromagnetic information transfer, including in antennas and waveguides (e.g., fiber-optics).
- Extended methods of calculating performance limits for photonic multi-material, non-linear optical, and cloaking devices.
- Developed an open-source package for calculating photonic performance limits. [GitHub repo.](#)
- Currently working on heat transfer formulations for limits on broadband thermophotovoltaics.

Undergraduate Research Fellow, Chan Group, Caltech

Researched tensor networks for simulating many-body quantum physics.

Summer
2020

Undergraduate Research Fellow, Gray Group, Caltech

Researched the ligand field theory and electronic structure of boronated cyanometallates via spectroscopic methods.

Summer
2019

Undergraduate Research Fellow, Gray Group, Caltech

Researched methods of electrochemical reduction of carbon dioxide to produce oxygen.

Summer
2018

CONFERENCE TALKS

Photonics North – Invited Talk

“Certificates of Suboptimality & Opportunities for New Photonic Designs”

2026

16th International Conference on Metamaterials, Photonic Crystals and Plasmonics – Invited Talk

“The photonic gearbox: ultra-high local density of states via cylindrical-symmetry breaking”

2026

JOURNAL REVIEWING

Journal of the Optical Society of America B
Computational and Applied Mathematics

ACS Photonics
Science Advances

TECHNOLOGIES

Proficient in Python (+NumPy, SciPy), scientific computing and ML in JAX, C/C++, MPI, OpenMP.

AWARDS

Princeton Chemistry Department David V. Milligan '62 Fellowship

2021-2022

- Awarded by faculty in recognition of academic excellence

Reed and Ruth Brantley Research Fellowship

Summer 2020

Doris S. Perpall SURF Speaking Competition semifinalist (top 10% of presentations)

2019

Donald Voet and Jerome Vinograd Research Fellowship

Summer 2019

Dr. and Mrs. Daniel C. Harris Research Fellowship

Summer 2018

TEACHING

Teaching Assistant, Electronic Circuit Design, Analysis and Implementation – Held two weekly labs on electronic circuits. Teaching reviews.	Spring 2026
Teaching Assistant, Robot Planning Meets Machine Learning – Held help sessions and graded theory and programming questions for a graduate course on planning in robotics. Reviews hidden (new course).	Fall 2025
Teaching Assistant, Optical and Quantum Electronics – Held help sessions and wrote questions for a graduate course on light-matter interactions for applications in electronics. Teaching reviews.	Fall 2024
Teaching Assistant, Electromagnetism for Engineers – Held two weekly labs focusing on applications of electromagnetism to engineering. Teaching reviews.	Spring 2024 Spring 2025
Teaching Assistant, Ecology – Held three weekly labs focusing on experimental techniques in ecology. Prepared midterm and final exam questions. Teaching reviews.	Fall 2023
Teaching Assistant, Electromagnetism – Held regular problem-solving sessions, office hours, and labs for a first-year advanced course on electromagnetism. Teaching reviews.	Spring 2023
Teaching Assistant, General Chemistry – Held three weekly recitations with a mix of lecturing and problem solving. Prepared midterm and final exam questions. Teaching reviews.	Fall 2022
Teaching Assistant, Statistical Mechanics – Held weekly recitations, office hours, and an exam review session for an undergraduate course in statistical mechanics. Teaching reviews.	Spring 2021
Teaching Assistant, Physical Chemistry – Held weekly recitations and graded problem sets for a course on quantum chemistry. Prepared notes for the second part of the course.	Fall 2019

Selected Student Teaching Feedback

- *Electromagnetism for Engineers*: “Alessio was extremely helpful, cared about our understanding of the material, and was a very fair and kind individual.”
- *Statistical Mechanics*: “Alessio was an amazing TA. He was always very kind in the way he answered questions without making me feel like it was a dumb question. One of the best physics TAs I've encountered.”
- *General Chemistry*: “I feel as if most of my understanding of the course came from Alessio and his inherent ability to teach a group of students effectively.”